

Arnold tongue of two coupled PING circuits

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Abstract

Reciprocal excitation should synchronize two PING circuits over a wider natural-frequency mismatch as coupling increases, while preserving the rule that the intrinsically faster circuit leads in phase [1]. We test that claim in a graph-native sweep over measured detuning and coupling. The 80 E / 20 I network recovers a centred Arnold tongue that widens across 4 successive coupling steps, and all 715 primary trials are valid. One near-resolution cell violates the strict phase-sign criterion, so the 80 E / 20 I reproduction failed despite 98.7% sign agreement. A focused 800 E / 200 I confirmation resolves that exception: all 40 coupled trials lock with the expected phase sign.

Methods

1. **Fix the graph and coupling intervention.** Each circuit contains 80 conductance-based excitatory neurons and 20 inhibitory neurons. A private 80-channel Poisson population drives each excitatory population. Local E-to-I AMPA and I-to-E GABA-A projections generate PING activity. Four reciprocal AMPA projections share coupling weight K : E_A projects to E_B and I_B , and E_B projects to E_A and I_A . The graph and network seed remain fixed. Figure 1 shows the intervention; the evidence-scale result verifies the realized parameter tensors.
2. **Calibrate natural frequency before coupling.** Equal-drive, $K = 0$ trials identify a stable monotonic input-rate interval. For each paired seed, the horizontal coordinate is the measured natural-frequency difference

$$\Delta f_0 = f_A^0 - f_B^0, \quad (1)$$

where Δf_0 is natural detuning, and f_A^0 and f_B^0 are the uncoupled gamma peak frequencies of circuits A and B. Input-rate difference is not used as a substitute. Figure 2 reports the accepted interval and its rejected boundary conditions.

3. **Measure phase and concentration.** Excitatory spikes are converted to population rates with a fixed 5 ms Gaussian kernel. Gamma frequency comes from the post-transient spectrum. A zero-phase 25–90 Hz band-pass filter and analytic signal define relative phase

$$\varphi(t) = \text{unwrap}(\theta_{A(t)} - \theta_{B(t)}), \quad (2)$$

where $\varphi(t)$ is unwrapped relative phase at time t , and $\theta_{A(t)}$ and $\theta_{B(t)}$ are instantaneous phases. Phase concentration is

$$R = \left| \frac{1}{T} \sum_{t=1}^T \exp(i\varphi(t)) \right|, \quad (3)$$

where R is phase-locking value (PLV), T is the number of analysed time samples, and i is the imaginary unit. Figure 4 maps these estimators; Figure 5 shows their time-domain behaviour.

4. **Freeze the locking rule.** Repeated uncoupled equal-drive trials set the admissible emergent frequency difference, relative-phase slope, and phase-slip count. A valid trial is locked when

$$L = \mathbf{1}[|f_A - f_B| \leq \varepsilon_f] \mathbf{1}[|a_\varphi| \leq \varepsilon_\varphi] \mathbf{1}[N_{\text{slip}} \leq \varepsilon_{\text{slip}}], \quad (4)$$

where L is the binary locking label, $\mathbf{1}[\cdot]$ is an indicator, f_A and f_B are coupled gamma frequencies, ε_f is the frozen frequency-difference tolerance, a_φ is the fitted slope of Equation 2, ε_φ is its tolerance, N_{slip} is the complete phase-slip count, and $\varepsilon_{\text{slip}}$ is its tolerance. PLV from Equation 3 is descriptive, not part of L . Figures 3 and 4 apply Equation 4 without post hoc threshold changes.

5. **Execute the frozen primary grid.** Thirteen target detunings cross eleven coupling levels with 5 paired input seeds per cell. Each trajectory lasts 3000 ms at a 0.1 ms timestep; the first 500 ms are discarded. A trial is invalid if recorded state is non-finite, any E or I firing rate is below 1 Hz, or either E population lacks a resolved 25–80 Hz peak. Figures 3–5 report this frozen grid.
6. **Apply the registered verdict.** Geometry passes only if the locked region is contiguous, centred near zero detuning, and widens across at least three successive nonzero coupling levels. Every grid cell must contain a valid trial. In every locked nonzero-detuning cell, the circuit with greater natural frequency in Equation 1 must lead in phase under Equation 2. The verdict subsection applies these clauses literally.
7. **Run a focused finite-size confirmation.** Because only one phase-sign cell fails at 80 E / 20 I, the follow-up increases each population tenfold and tests mirrored target detunings at the uncoupled, disputed, and stronger coupling levels. It retains E-population spikes and summary observables, uses ten paired seeds per cell, and lengthens the post-transient window to 5 s. Figure 6 reports the result without reopening the primary grid.

The evidence archive bit-packs exact input and population spikes for every primary and follow-up cell. The primary archive also retains population-mean voltage and projection conductance at 1 ms resolution, compiled bundles, input hashes, seed ledger, and one realized parameter set per coupling level.

Results

The authored graph isolates reciprocal coupling

Method 1 varies only the four cross-circuit AMPA projections. The local PING loops, private drives, graph topology, and seed are fixed, so a change across K is attributable to the registered coupling intervention.

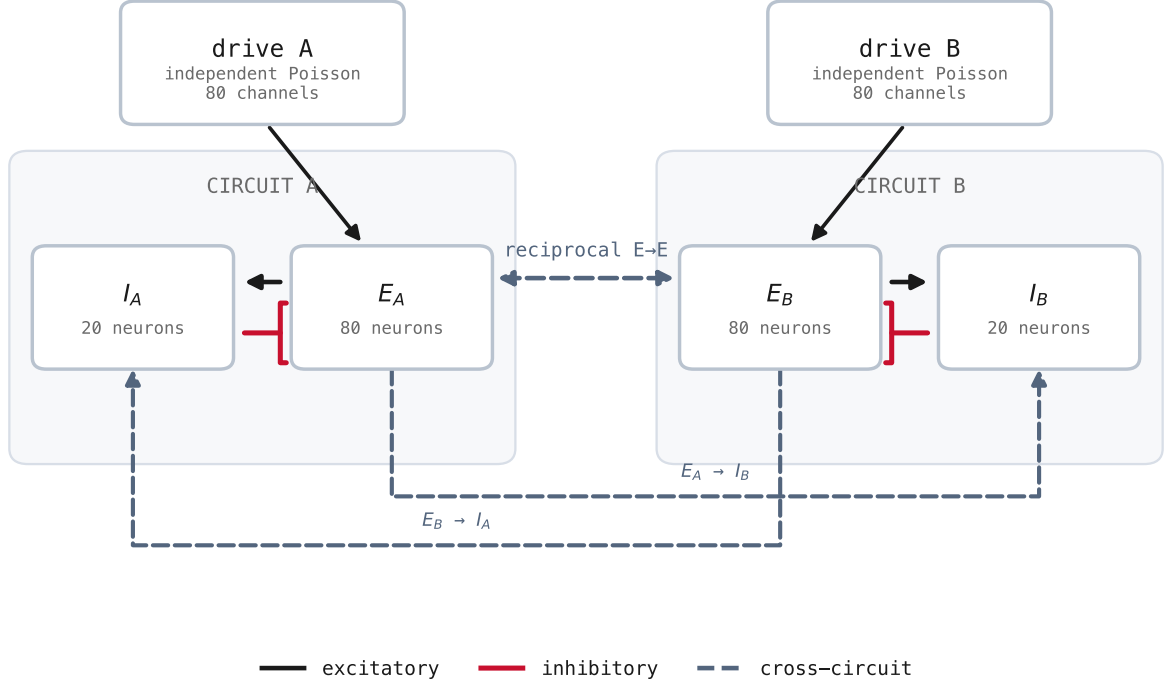


Figure 1: Two-circuit coupling intervention. Each private Poisson input drives a local E-to-I-to-E PING loop. Dashed pathways are the four reciprocal AMPA projections varied together by coupling weight K ; red bar-headed pathways are local GABA-A inhibition. Only the dashed pathways vary across the primary grid.

The realized parameter archive confirms that all non-coupling tensors remain invariant across 11 coupling levels, as required by Method 1.

Calibration defines a stable detuning axis

Method 2 yields a stable monotonic interval from 70 to 130 Hz per input channel. This interval supplies the rate pairs used to evaluate Equation 1 throughout the primary map; their measured natural detunings closely track their targets.

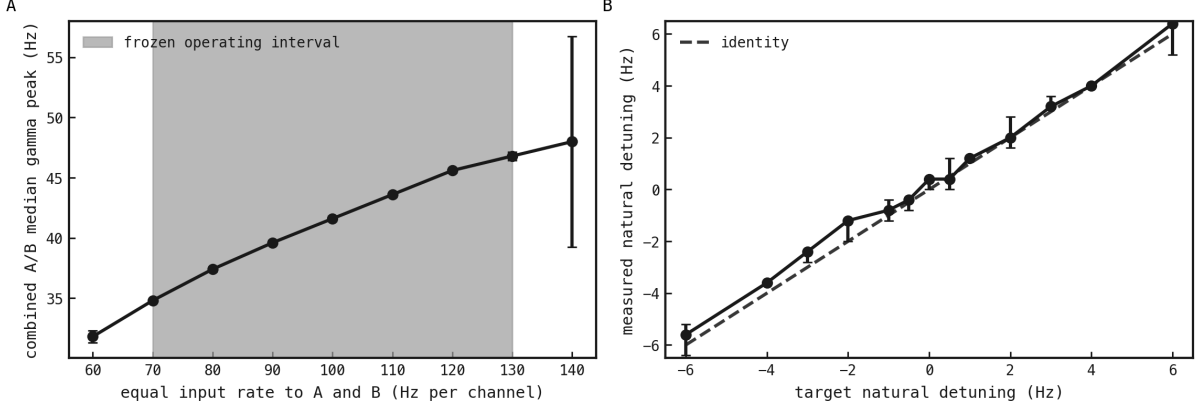


Figure 2: Calibration of frequency and natural detuning. (A) Circuits A and B receive the same input rate. Each point is the median of 10 gamma-peak estimates: both circuits from each of five trials. Bars show half the interquartile range of the combined estimates. The shaded 70–130 Hz interval supplies the rate pairs used in Equation 1. The lowest-rate point exceeds the registered within-rate IQR ceiling. At the highest rate, the estimator hops between two separated spectral modes across seeds, producing the large bar. Both boundary conditions are excluded. (B) Equation 1 defines the measured uncoupled natural detuning. Each point compares its median across five paired seeds with the target; bars span the interquartile range, and the dashed identity line marks exact realization.

The frozen thresholds used in Equation 4 are 0.8 Hz for frequency difference, 11.53 rad/s for absolute phase slope, and 4 complete slips. The grid is identified by checksum 8813df0395d0aeb00eb4613d28f776cc09cb4ff783694a7c7603e749c79f3c0f.

Coupling produces a widening Arnold tongue

Applying Equation 4 to the frozen grid from Method 5 produces a contiguous locked region centred near zero detuning. Its width reaches 11.92 Hz at the largest coupling and increases across 4 successive coupling steps.

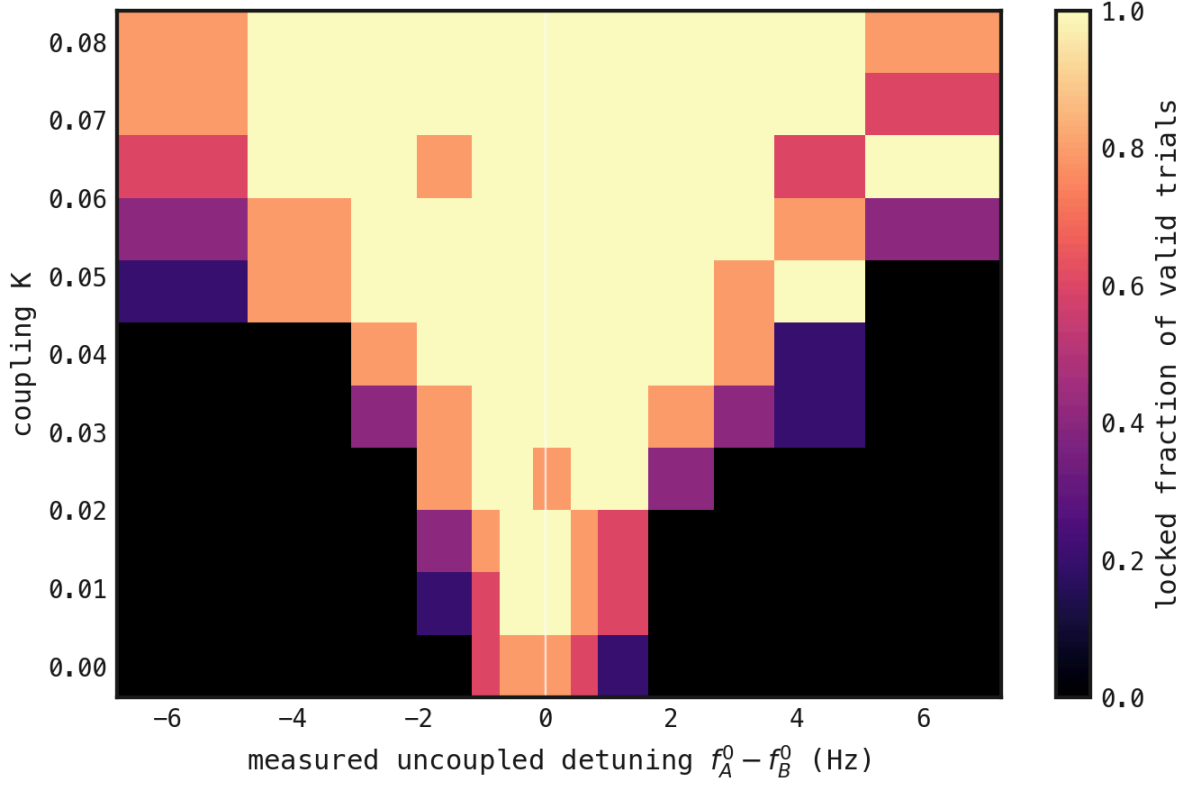


Figure 3: Primary Arnold tongue. Measured natural detuning Δf_0 from Equation 1 is on the horizontal axis in hertz; reciprocal coupling K is on the vertical axis; colour is the fraction of valid trials that satisfy Equation 4. The centred locked region widens with coupling, satisfying the registered geometric clause.

The tongue is not an artefact of invalid or silent trials. All 715 primary trials are valid, and the component estimators from Methods 3 and 4 change coherently across the same grid.

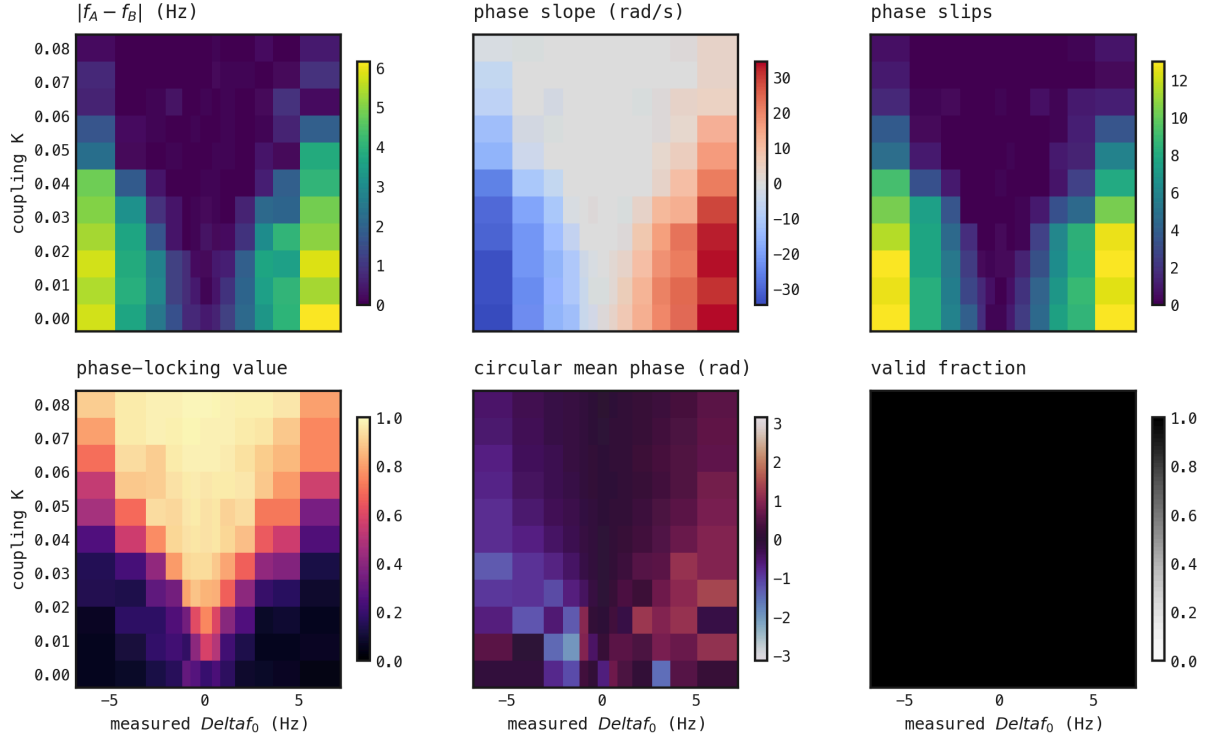


Figure 4: Component estimators on the primary grid. All panels share measured natural detuning in hertz horizontally and coupling K vertically. Panels report coupled frequency difference in hertz, fitted slope of Equation 2 in radians per second, complete phase slips, PLV R from Equation 3, circular mean phase in radians, and valid-trial fraction. Validity is 100% across the grid, while low frequency difference and phase drift coincide inside the Arnold tongue.

Representative conditions selected before inspection show the same transition in time: phase remains bounded inside the tongue and drifts at the immediately preceding coupling.

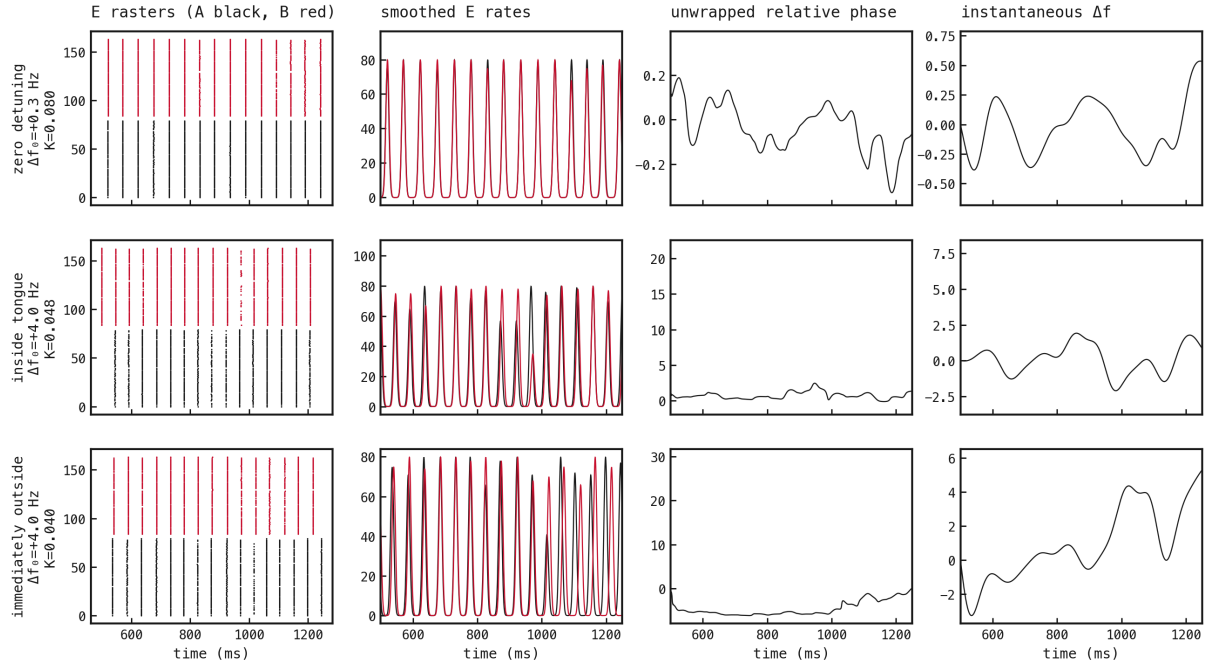


Figure 5: Predetermined time-domain checks. Rows show zero detuning at maximum coupling, the first locked positive-detuning condition, and the same detuning at the immediately preceding coupling. Columns show E rasters, 5 ms-smoothed E rates in hertz, unwrapped relative phase $\varphi(t)$ from Equation 2 in radians, and instantaneous frequency difference in hertz. Relative phase is bounded inside the registered tongue and drifts immediately outside it.

The strict 80 E / 20 I verdict fails one phase-sign cell

Method 6 gives a split result. Geometry and validity pass, but one qualifying cell has the wrong phase sign. Agreement is 98.7%, while the registered clause requires every qualifying cell to agree. The overall 80 E / 20 I reproduction therefore failed. The exception lies near the estimator's registered resolution, so Method 7 tests that boundary rather than repeating the full grid.

The 800 E / 200 I confirmation resolves the exception

The focused confirmation passes: all 40 coupled trials are valid and locked under Equation 4, and all 40 have the phase sign required by Equation 1 and Method 6. At the disputed negative-detuning cell, all 10 seeds have the correct sign with mean phase -0.543 rad. The mirrored positive condition gives 0.526 rad.

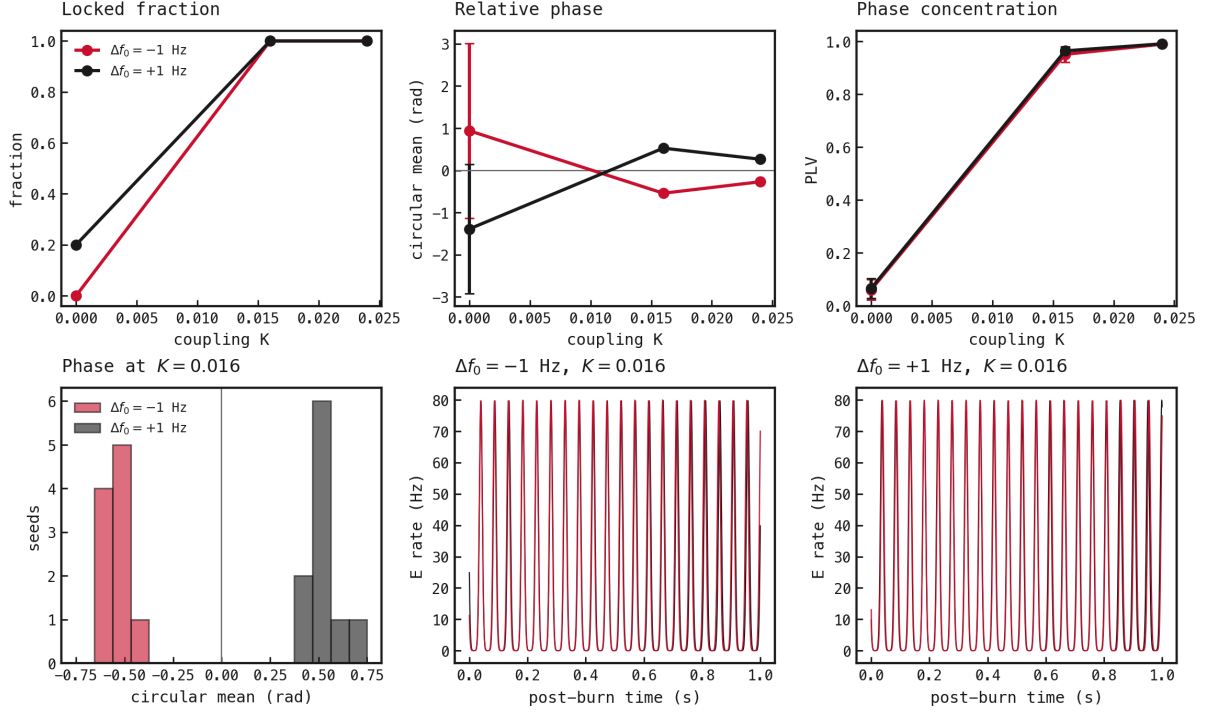


Figure 6: Focused 800 E / 200 I finite-size confirmation. Top panels show locked fraction, circular mean relative phase in radians, and PLV R from Equation 3 over coupling K for mirrored natural detunings; error bars are one standard deviation across 10 paired seeds. The lower-left panel isolates phase at the disputed coupling. The remaining panels show post-burn E-population rates in hertz (circuit A black, circuit B red). Both detuning signs lock with mirrored phase ordering, resolving the lone 80 E / 20 I exception.

The follow-up supports finite-size or estimator-resolution noise, rather than a systematic phase-ordering contradiction, as the cause of the original exception.

Evidence scale

Dense primary retention would occupy 27.82 GB. The published exact-event and state-summary archive occupies 165.5 MB, a 168.1-fold reduction. The 143 primary cells required a median 19.03 s each locally.

References

1. Lowet, Roberts, Hadjipapas, Peter, van der Eerden & De Weerd (2015): *Input-Dependent Frequency Modulation of Cortical Gamma Oscillations Shapes Spatial Synchronization and Enables Phase Coding*. PLOS Computational Biology. doi:10.1371/journal.pcbi.1004072